City charges for competing delivery carriers: exact best responses under probability-box travel times and leader pricing without equilibrium search

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Abstract

A city that charges delivery carriers faces a bilevel problem with three stacked difficulties. Each carrier solves an NP-hard routing problem under probability-box travel times with unknown dependence; the carriers play a Cournot-type congestion game; and the usual search over charges solves one equilibrium per grid point, about two thousand routing solves per city. We give an algorithm for each layer. Budget-resource labelling prices routes exactly, and branch-and-price makes each carrier's integer plan exact; it proves optimality with 22 customer zones in all nine cities and 30 to 40 customers in London and New York, with far fewer labels than scenario labelling, and probability boxes buy a guaranteed upper bound and a lower overtime risk at a higher expected cost than sample averages. Dual implementation pricing makes a social target an equilibrium through best-response cuts. Aggregate-response pricing sets charges from observed zone totals alone and contracts at a rate independent of instance size. On nine metropolitan areas built from OpenStreetMap and gridded population, with ten seeds in three cities, dual implementation pricing reached 1.07–1.23 times the best grid point's gain with 5.7–8.3% of its calls, and 1.00–1.02 of the social optimum's gain (grid: 0.815–0.938); aggregate-response pricing reached 0.933–0.995. Pure equilibria exist under exposure-based congestion, and deciding their existence is strongly NP-complete under vehicle-count congestion; the belief-width result builds on earlier work on approximate equilibria. Levels of the price of anarchy, service and gains depend on the criterion the carriers use, whereas orderings and policy directions do not. Instances and code are released.

Keywords

urban logistics; bilevel optimization; vehicle routing; imprecise probabilities; congestion games; city charges

Word count

About 13,700 words of main text (title page to the end of the conclusion, including abstract, headings, equations and table text; excluding the declarations and references).

Length and format

Main manuscript: 34 pages including abstract, figures, tables and references (limit 35); A4, single column, 11 pt, 1.5 line spacing, no line numbers. Supplementary Material (separate file): 50 pages.

Previous submission

This is a resubmission based on the previously rejected paper OMEGA-D-26-03236 (decision: reject & resubmit); see the cover letter.

Data and code availability

Code, processed data, parameter sources, result files, figures and the table-generating scripts are available at https://github.com/vinhqdang/Competing-Carriers-under-Uncertainty.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRediT authorship contribution statement

Quang-Vinh Dang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing - original draft, Writing - review & editing, Visualization, Project administration.